1 Lecture 10 Interpretation of OLS Short Answer Questions

2 Question #1

The OLS estimator is

$$b = (X'X)^{-1}X'Y,$$

where X is $n \times K$, Y is $n \times 1$, and b is $K \times 1$.

Define the prediction error associated with the OLS coefficients as

$$e_i = y_i - x_i'b.$$

Let H denote the $n \times 1$ vector of OLS prediction errors:

$$H = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}.$$

a. Define the projection matrix for X, P_X.

b. Show that

$$\mathrm{P}_X H = 0.$$

c. State the result from part (b) in words.

d. Consider predicting the OLS errors using X:

$$\hat{e}_i = x_{i1}a_1 + x_{i2}a_2 + \cdots + x_{iK}a_K.$$

Show that the OLS coefficient vector a is equal to zero.

3 Question #2

Let Y, Z, X, and $\boldsymbol{\varepsilon}$ be $n \times 1$ vectors related by

$$Y = \tau Z + \kappa X + \boldsymbol{\varepsilon}.$$

Suppose every element of Z is equal to 1. For context, let Y denote total sales across n observations, X the price at each observation, and $\boldsymbol{\varepsilon}$ all other factors affecting sales.

a. Define the projection matrix for Z as

$$P_Z = Z(Z'Z)^{-1}Z'.$$

Describe P_Z completely. What are its dimensions, and what is the value of every element of the matrix?

b. Consider the matrix product $P_Z X$. Describe the result completely. What are its dimensions, and what is the value of every element of the resulting vector?

c. Define the residual-maker matrix for Z as

$$M_Z = I - Z(Z'Z)^{-1}Z'.$$

Multiplying the model by M_Z gives

$$Y^* = \kappa X^* + \varepsilon^*,$$

Describe X^* in this equation completely. What are its dimensions, and what is the value of every element of the vector?